

Homework 3

1. Consider the autoconvolution operator K in $L^2(0, 1)$ defined for $x \in L^2(0, 1)$ by

$$K(x)(t) = \int_0^t x(t-s)x(s) \, ds, \quad s \in (0, 1)$$

with linearization around $x_0 \in L^2(0, 1)$ given by

$$K'(x_0)h(t) = 2 \int_0^t x_0(t-s)h(s) \, ds, \quad t \in (0, 1).$$

- (a) Take $x(s) = 2e^{-2s}$ and show that $y(t) = K(x)(t) = 4te^{-2t}$.
 - (b) Implement K numerically using a quadrature rule as a mapping from $\mathbb{R}^{256}$ to $\mathbb{R}^{256}$. Use your code to find an approximation of $K(x)(t)$ and compare graphically to $y(t)$.
 - (c) Add noise of relative size 5% to y to get y_δ and consider solving $K(x) = y_\delta$ by linearization: For the constant function $x_0(t) = 2 - 1.5t$, discretize the operator $K'(x_0)$ as a matrix $A_0 \in \mathbb{R}^{256 \times 256}$. Solve using Tikhonov regularization the linearized and discretized inverse problem to find an approximate solution $x_1(t)$. Choose the regularization according to one of the parameter choice rules. Plot the solution and compare to $x(t)$. Calculate the error.
 - (d) Take another Newton step to find $x_2(t)$.
 - (e) What happens if you start from the constant function $x_0(t) = -2 + 1.5t$ instead?
 - (f) Is the inverse problem ($K(x) = y$) ill-posed; is it locally ill-posed?
2. The radial EIT/Calderón problem linearized around $\gamma_0 = 1$ is defined through the operator K as follows

$$Kh(n) = -2 \int_0^1 h(r)r^{2n-1} \, dr = \lambda_n, \quad (1)$$

with λ_n denoting eigenvalues of $\Lambda_\gamma - \Lambda_1$.

- (a) In the file `bHW3.mat` on inside you find 10 precomputed ('measured') eigenvalues for a particular $\Lambda_\gamma - \Lambda_1$. Compute an estimate of $\gamma = 1 + h$ using both Tikhonov and TV regularization and plot it.
- (b) Knowing that γ has the radial form

$$\gamma(x) = \begin{cases} 1 + \hat{q}, & |x| \leq R, \\ 1, & R < |x| \leq 1. \end{cases}$$

give an estimate from the above of the contrast $\hat{q}$ and radius R .

Important notes:

- Deadline for the HW3 is Monday December 9 at 12:00.
- All homeworks must be approved in order to qualify for the exam.
- Your background is different, and therefore you can choose to put emphasis differently in your solution.
- You must demonstrate your understanding of the material.
- Each student should hand in a single pdf via Learn. The number of pages for the main report must not exceed 5. Code can be put in an appendix.
- Comment on all results: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions.